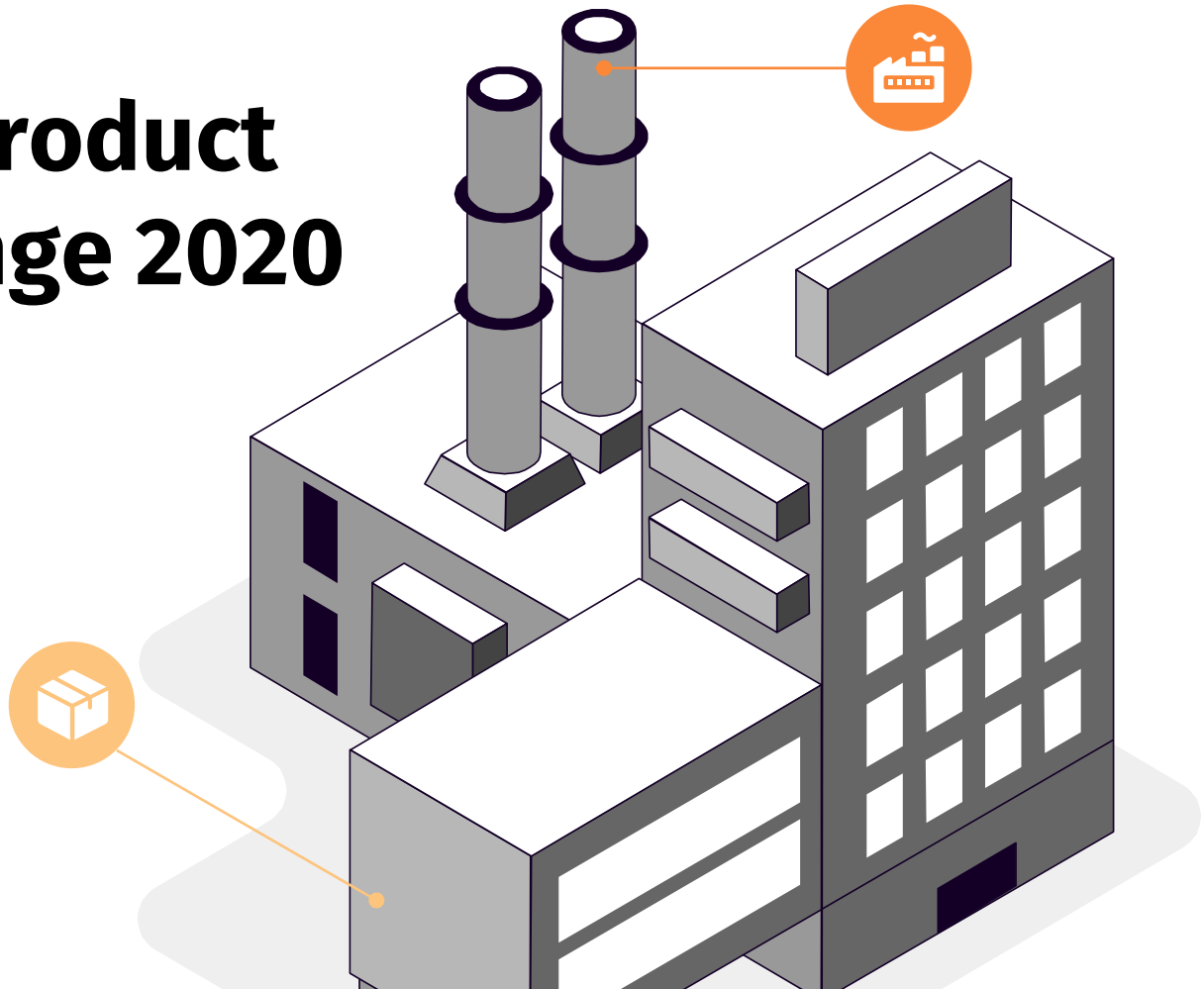
Honeywell Product Case Challenge 2020

Team Name: Skyrocketeers
Theme: Industrial Worker

Ayush Jain
Niharika Sharma
Sai Keshav Gaba
Sai Kiran Wupadrasta

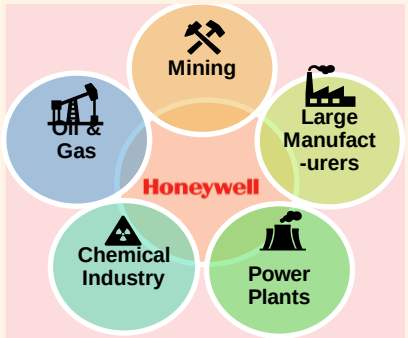


Executive Summary

Selected theme: Industrial Worker

Reason for choosing the theme:
The team consists of members that come from Electrical engineering, automobile, and software industry background. We relate to the potential growth and challenges faced by stakeholders in the industry and believe that innovative technology can solve many industry problems.

Customer Segments



[1]On the basis of market size, turnover, growth rate in India, openness to adopt software solution, & size of the potential and current customers of Honeywell

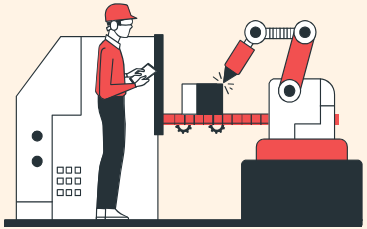
Executive Summary:
Surveying 19 stakeholders across the mining, oil & gas, power plants, and chemical industries, we identified the critical problems of raising, tracking, monitoring, and recording faults across systems. To solve this challenge, we present a software-based fault management system that provides ease of pointing out faults using Image processing, automatically creates incidents and notifies across the siloed teams, and helps track and document incidents at the single source of truth. The data collected is then used to create a centralized dashboard to provide trend analysis, streamlined communication, and predictive analysis.

Problems being targeted:
- **Connectivity:** When an issue is raised it takes a long time for stakeholders to communicate between the siloed teams and the workers.
- **Paper – based documentation:** Inspectors, or engineer either physically align to communicate a fault or rely on paper-based solution, checklist, working instructions, etc. are all done on papers and result in huge pile of paper records that cannot be accessed or utilized for references
- **Siloed teams:** Because of face to face on the field interaction and paper-based documentation, sharing information between teams is difficult. Hence the teams keep working in silos and are not able to track the records or access a single source of truth
- **No analytics possible on historical data:** Due to difficulty in accessing paper-based records, workers end up redoing the work and rely on human-memory and personal learnings to apply for future references

*Refer Exhibit 4 for Industry Analysis

User Personas and Problem Identification

Surveying 19 stakeholders across the mining, oil & gas, power plants, and chemical industries, and reviewing secondary research, we identified three personas and their needs. The problems raised were in areas of raising, tracking, monitoring, and recording faults across systems.



Inspection Manager

Jagdeep: Age: 37 years | Work Ex : 15 years

Responsibilities

- Conduct tests to check quality of products
- Monitor production processes
- Identify malfunctions
- Create checklists of observations
- Raise issues for the identified anomalies

Key Problems

- SOPs and Standard **work instructions are paper based, complex and hard to read**
- **Checklist creation** & critical inspection steps are **paper based**
- **Difficult to communicate details of** a malfunction to the relevant stakeholder without media reference



Maintenance / Line Engineer

Charan: Age: 30 years | Work Ex : 7 years

Responsibilities

- Reviewing health & safety performance outcomes
- Supervise and fix the fault raised
- Inform stakeholders if further actions needed
- Analysis of repetitive equipment failures

Key Problems

- **Track and record issues raised**, and fixes made
- **Quickly spot the area of problem** on the field
- **Track a malfunction without media reference**
- Delay in **connecting** and aligning with different stakeholders such as electrical engineer, mining surveyor/inspector, department head
- No data analytics available to hint high probability area of fault

Bharati: Age: 40 years | Work Ex : 17 years

Responsibilities

- Analyzing production data to resolve & identify production issues
- Ensure performance management of the plant
- Interact with different managers and track issue handling
- Raise a request to schedule an inspection



Department Head

Key Problems

- **Track status of issues raised**
- **Unavailability of single source of truth** due to paper-based documentation
- **No trend analysis** due to paper-based documentation
- **Delay in connecting, communicating and aligning** with different stakeholders such as quality control team, inspection managers etc.

Solution: Fault Management System

Paper Based Processes | Siloed Systems | Asset Tracking

We prioritized the problems that had the maximum impact on the stakeholders. After multiple rounds of discussions along with secondary research, we designed a 1 stop solution to manage the extensive inspections and the vast array of faults and issues routinely identified in the plant. Various stakeholders would collaborate using the software for easy tracking and quick resolution. The following features are suggested basis RICE prioritization:

Inspection

Digi-Inspect

Software will use image processing of camera feeds to perform visual inspections and analyze sensor data input to monitor control parameters

Digital Checklist

A customised digital checklist will be provided for manual inspection and data input would be possible using on-device touch

Schedule Inspection

Department head has provision to schedule and assign an ad-hoc inspection task

Inspection Dashboards

A curated dashboard basis customer requirements to be provided to track the historical inspection records, completion status, inspection issue trends, etc.

Recommendation System

Recommendation on high priority areas prone to frequent failures will be generated using ML to arrest issue before occurrence

Digital Incident Record

Fault Tagging

- Member will use the device camera to capture the fault location on AR software tool and tag it to a predetermined category or annotate on the screen. This would create a new instance in the software, tagged to the QA department.
- The QA engineer can then access the annotation by hovering the device camera over the same location, leveraging AR capabilities. The AR capabilities will help locate the exact place of fault in a complex and challenging environment.

Alert Management

The engineer can raise an alert to the department in case of an high priority issue for a quick redressal

Worker Traceability

Bluetooth Low Energy wireless communication design allows handheld computer devices to communicate with each other at low range while preserving battery life.
Incident records are shared across devices and stored on in-house servers for further viewing on manager dashboard

Speech-to-text comments

Member can speak into the device explaining the problem in detail which will be added as textual comments for reference

Fault Management

Incident management

A dashboard solution provided with customized metrics to track the incidents raised and collaborate with croff functional team towards successful closure
Ability to assign to respective stakeholders, record their inputs and developments, and track the overall progress and health of the plant

Remote Assistance

Field Engineer can be assisted by a technical expert remotely on the CT60 devices. Guidance can be provided using annotation sharing

Smart Recommendations

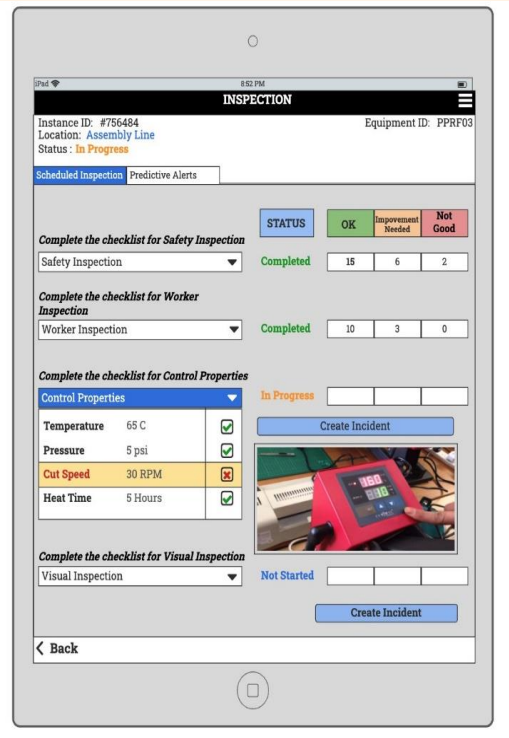
Dashboard has the ability to learn from the recurring issues using pattern match and generate intelligent recommendations on the possible root causes and error proofing methods (Poka-Yoke)

Central repository

Structured data repository that can be accessed to reference any historic data and can be used for training new members on the past issues

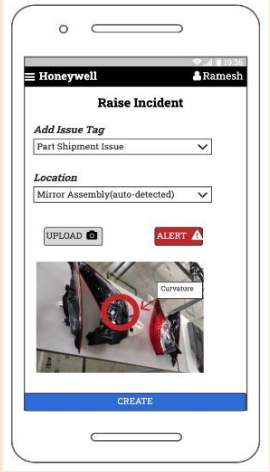
Fault Management System: Product Wireframes

Inspection



Checksheets are digitized and the auditor can create an incident to report an anomalies

Digital Incident Record

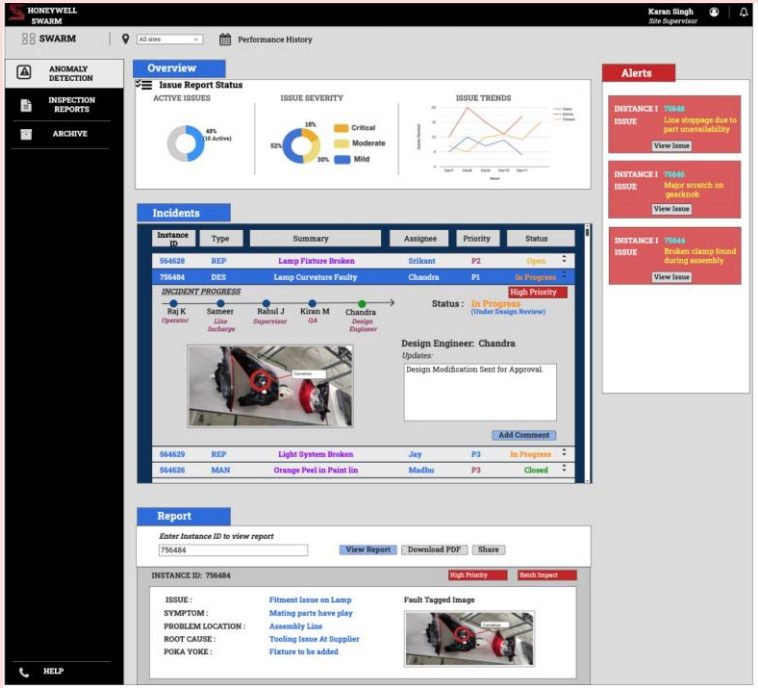


Operators can use this interface to create an incident by uploading the annotated AR-based scan and tagging the incident to the issue



This is how the AR based scan upload and reference will work

Fault Management



This is the dashboard visible to the department heads to monitor, assign, track, and update the fault progress. It also gives recommendations on the root cause of the problem and the potential error proofing

Roadmap and Timelines

Design Application Phase 1	Development – MVP Phase 1	App Launch Phase 1	Development – Feature Enhancements Phase 2	App Enhancements and roll out Phase 2
Compare and align the planned design with the voice of the customer and create a customer requirement document	Focus on implementing key features from the RICE framework for the minimum viable product	Launch the application on CT60's integrating with existing features	Enhance application features by implementing features from the RICE framework	Implement the feature changes in the application. Roll-out latest features in the existing software application and prepare for collecting data

Timeline		Month 1-3		Month 4-12		Month 13-18		Month 19-24		Month 25-30	
Phases		*Phase 1*						*Phase 2*			
Milestones		Application Design		Development MVP		Launch App		Feature Enhancements		Feature Migration	
Teams	Roles										
Analytics	Modelling	Metrics and Data Analysis	AB Testing	Metrics and Data Analysis	AB Testing	Metrics and Data Analysis	AB Testing	Metrics and Data Analysis	AB Testing	Metrics and Data Analysis	AB Testing
Development	Infrastructure	Database Streamlining		Database Streamlining		Database Streamlining		Database Streamlining		Database Streamlining	
	Backend		DB Integration/Development Design		DB Integration/Development		DB Integration/Development		DB Integration/Development		DB Integration/Development
	Frontend	Frontend Designing		Development(App Features)		Development (Application Feedback)		Development(Enhancements)		Development (New features)	
	QA		Variance/System/AB Testing		Variance/System/AB Testing		Variance/System/AB Testing		Variance/System/AB Testing		Variance/System/AB Testing
UI/UX	Design	Wireframing/UI Design		Wireframing/UI Design		Wireframing/UI Design		Wireframing/UI Design		Wireframing/UI Design	

Go-To-Market Strategy

	Attract	Convert	Close	Delight
Goals	> Demand generation > Brand Awareness	> Lead generation	> Negative Churn > Expanding Revenue	> Retention > Monetization > Customer Success
Strategies	> Promote company's customer first image > Exhibitions and conferences > Trade Shows > Webinars > Direct mail	> Cross Selling – Promote sale along with existing Honeywell solutions > Direct selling – Sales Personnel presents product features > Government collaboration to subsidize for MSMEs > Sales promotions- Discount price based on sales volume and brand loyalty > Pricing based on factory/plant size > Subscription based model	> Point of sales through Honeywell partner vendors > Direct Selling > Honeywell website (license renewal or software upgrades)	> License Renewals > Feature Upgrades > After sales support > Solution Deployment support > Software Training
Metrics	> Top channels > Number of meetings > Impressions	> Product qualified leads > Conversion rate to customer > Sales cycle > Customer acquisition cost	> Churn Rate > Customer lifetime value > New ARR and Expansion ARR > Annual revenue generated per feature >Average revenue per account	> MAU > Cost to service > Net Promoter Score > Number of support tickets created > Voice of customer data

Financial Projections



Key Assumptions

Growth rates

- Market share 20% pa
- Software Expenses 15% pa
- Hardware Expenses 10% pa
- Other expenses 15% pa
- No. of factories 2% pa

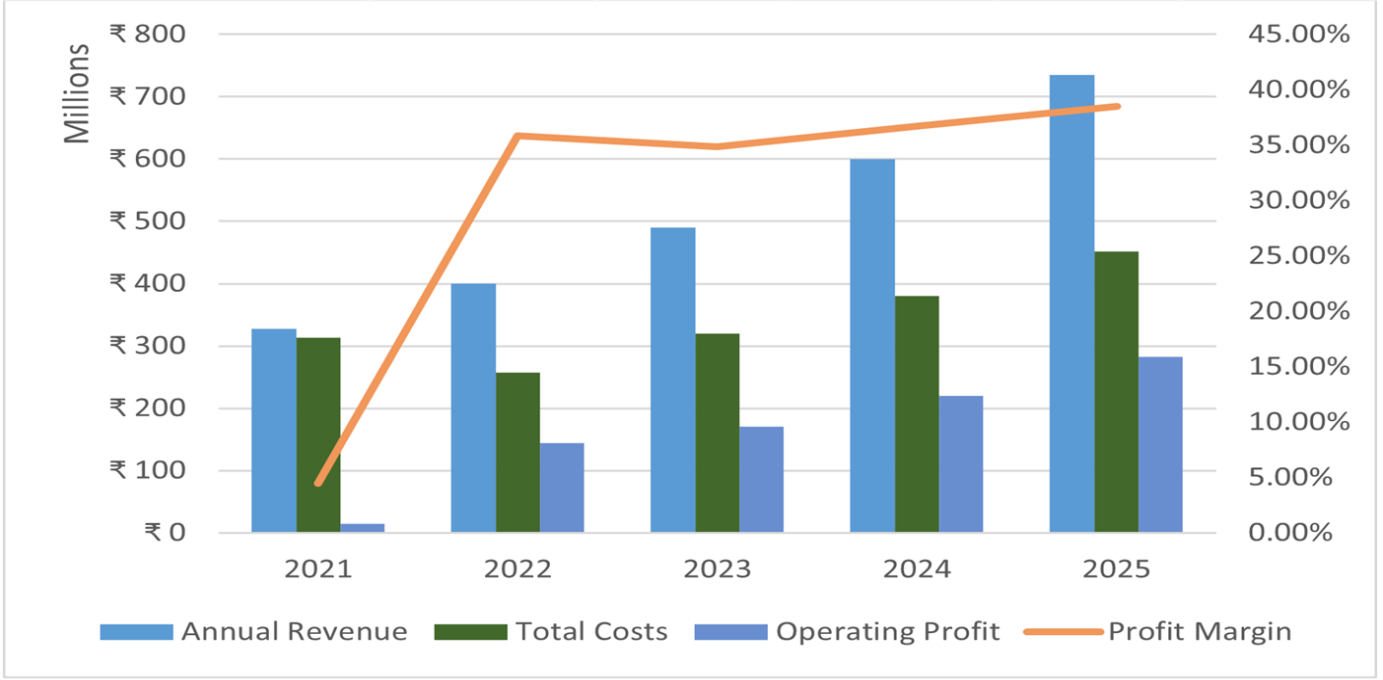
No. of units

- Mining sites 3100
- Oil & Gas 25
- Large Manufacturing 15000
- Large Power Plants 600
- Chemical Factories 14000

Expenses related

- Inflation 5% pa
- Year 1 hardware, software R&D 5+4 Cr
- Year 1 software R&D 4 Cr
- App & Dashboard Development Costs: Year 1: 4 Cr, Year 2: 2 Cr. 15% yoy rise
- Hardware Development Cost 2Cr once

Year	2021	2022	2023	2024	2025
Annual Revenue	₹ 32,72,50,000	₹ 40,05,54,000	₹ 49,02,78,096	₹ 60,01,00,390	₹ 73,45,22,877
Total Costs	₹ 31,27,12,500	₹ 25,68,91,470	₹ 31,94,55,813	₹ 37,97,51,944	₹ 45,17,18,549
Operating Profit	₹ 1,45,37,500	₹ 14,36,62,530	₹ 17,08,22,284	₹ 22,03,48,446	₹ 28,28,04,328
Profit Margin	4.44%	35.87%	34.84%	36.72%	38.50%



*For detailed calculations refer Exhibit 2

PRFAQ

Honeywell releases Swarm to digitize industry workspace operations

Swarm software solution to leverage augmented reality and machine learning capabilities to digitize fault management on industry workspace.

Honeywell is excited to announce the addition of Swarm to its product range of software solutions. Swarm uses augmented reality and machine learning capabilities to streamline fault record creation and management for the industrial workforce, track critical assets, & digitize the auditing process.
Connecting and aligning various dispersed stakeholders such as contractors, inspection managers, and factory floor workers is complicated. Also, most auditing procedures and SOPs are complex and paper-based, making the information less accessible.

With the integration of AR and ML capabilities, Swarm users will record industry workspace faults with visual notes and create automated fault tickets. Inspection managers and maintenance engineers will benefit from real-time access to critical data to prioritize and solve problems more quickly. Managers can use the analytics-enabled Swarm dashboard to derive significant insights to save time and costs through platform recommended solutions.

"*Swarm will assist our customers in making fault management smooth and efficient. With Swarm, we are looking to make traditional paper-heavy industries more digital in the way they work*," said Niharika Sharma, Sr. Product Manager at Honeywell.

The software is easy to use and Honeywell provides training and deployment support for quick ramp up. For industry workers, the software allows for note making and is also available in vernacular languages.

"*The Swarm software has been beneficial to our inspection managers to track workflows & check fault reports. It has helped factory workers to capture the issue and make notes easily,*" said Mohit Kapoor, Chief Technology Officer at Mahindra Group.

Honeywell Swarm is now available for sale with all partner vendors. To learn more, log on

to https://sps.honeywell.com/us/en/software/productivity/workforce-task-management

FAQs

External:

Can the product be incorporated with internal DB management systems?
Swarm can be incorporated with existing DB servers deployed at the firm & Honeywell cloud. During network issues, the data can be synchronized once connectivity is restored.

Can the software be used on existing handheld computer solutions?
Specific licenses & software versions are needed to enable the software on existing Honeywell devices.

Is it a scalable solution if we want to add more industry floor size?
Swarm supports up to 5000 handheld computers & provides coverage to large industry workspaces.

Is the dashboard customizable by industries?
Each dashboard & audit screen can be modified to track a host of KPIs & checklists by industry.

Internal:

What will customers like most about Swarm?
AR will help ease to locate issues maintenance and help pointing out the exact problems.

What will customers like the least about Swarm?
There may be a steep learning curve for inspection managers who may not be as technically inclined.

What may be the opportunities for Swarm?
Pandemic will give thrust to manufacturing & digital tracking
Industrial reforms and increased govt incentives for MSMEs
Greater penetration of cellular networks may improve opportunities to connect Swarm & Honeywell Cloud

How do we compare to our competitors?
We are the first to introduce AR technology for inspection management, and our capabilities match industry competitors such as Bosch and Siemens in digital twin and analytics.

Elevator Pitch



Mining

For mining firms, who face complexities in tracking and communicating with workers, Honeywell Swarm is a worker tracking solution that provides last known location as well as speech to text provision for workers to communicate back in a contactless and handsfree way. Unlike our competitors such as Siemen's SIMATIC, we provide capability inbuilt with software with no requirement for additional sensor tags.

Power, Oil and Gas

For power, oil and gas firms, who face remote connectivity issues for reporting field issues, Honeywell Swarm is a communication solution that provides Low Energy Bluetooth based communication as well as remote assistance feature for seamless incident ticket tracking and resolution. Unlike BLE solutions from competitors such as Bosch, we provide high level of scalability to cover close to 5000 device connections.



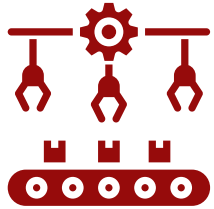


Chemicals

For chemical firms, who are plagued with improper maintenance and late fault detection, Honeywell Swarm is a fault management system that provide quick and easy fault recording enabled by augmented reality and a digital record keeping platform. No competitor currently provides augmented reality capabilities.

Manufacturing

For manufacturing firms, for whom machine checks and repairs are complex and time consuming, Honeywell Swarm is a tracking solution that automates incident and audit record creation and aligns all stakeholders. Unlike our competitors such as Siemen's Industry software, we provide high level of predictive as well as prescriptive analytics for better customer experience.



Product in a Box



HONEYWELL SWARM

Honeywell India Pvt. Ltd.

Feature-based subscription model from Rs.15,00,000/- onwards

In compliance with latest GDPR Policies, SOC 1/2/3, HIPAA, GLBA, FISMA & ISO/IEC 27000

Fault Management System



Swarm promises to save your Precious time and effort spent towards fault management by aligning all stakeholders and increasing overall efficiency

Contents of the box:

- Handheld Device Software
- Manager Dashboard Software
- Camera and Display Hardware
- License Renewals
- Feature Upgrades
- After sales support
- Solution Deployment support
- Software Training

Directions to use the product

> The engineers can use the CT60 device to scan and create an fault incident

> The supervisor and managers can track the progress of anomalies using the fault management dashboard

> Inspection manager can use the smart ‘digi-inspect’ application on the device

> Inspections can be tracked using the Inspection dashboard

Exhibits

Exhibit-1 : Market Sizing

Mining Sites	3100
Oil and Gas	25
Large Manufacturing (> INR 100 Cr)	15000
Large Power Plants	600
Chemical Factories	14000
Total	**32725**



Exhibit–2 : Financial Analysis and Forecast

Year	2021	2022	2023	2024	2025
Total Market Potential (no. of factories in segments)	32725	33379.5	34047.09	34728.0318	35422.59244
Target share of Market Potential (%)	0.50%	0.60%	0.72%	0.86%	1.04%
Target share of Market Potential (no. of factories)	163.625	200.277	245.139048	300.0501948	367.2614384
Average Solution Pricing (subscription per year per factory)	₹ 20,00,000.00	₹ 20,00,000.00	₹ 20,00,000.00	₹ 20,00,000.00	₹ 20,00,000.00
Annual Revenue	₹ 32,72,50,000.00	₹ 40,05,54,000.00	₹ 49,02,78,096.00	₹ 60,01,00,389.50	₹ 73,45,22,876.75
Application and Dashboard Development Cost	₹ 4,00,00,000.00	₹ 2,00,00,000.00	₹ 2,30,00,000.00	₹ 2,64,50,000.00	₹ 3,04,17,500.00
Application and Dashboard Maintenance Cost	₹ 2,00,00,000.00	₹ 2,30,00,000.00	₹ 2,64,50,000.00	₹ 3,04,17,500.00	₹ 3,49,80,125.00
Hardware Manufacturing & Development Cost	₹ 11,81,75,000.00	₹ 13,21,82,820.00	₹ 17,44,81,322.40	₹ 21,11,22,400.10	₹ 25,54,58,104.13
Support Services	₹ 2,00,00,000.00	₹ 2,30,00,000.00	₹ 2,64,50,000.00	₹ 3,04,17,500.00	₹ 3,49,80,125.00
Deployment and Training Costs	₹ 8,18,12,500.00	₹ 2,10,74,900.00	₹ 2,57,95,677.60	₹ 3,15,73,909.38	₹ 3,86,46,465.08
B2B Sales and Marketing Expenses	₹ 3,27,25,000.00	₹ 3,76,33,750.00	₹ 4,32,78,812.50	₹ 4,97,70,634.38	₹ 5,72,36,229.53
Total Costs	₹ 31,27,12,500.00	₹ 25,68,91,470.00	₹ 31,94,55,812.50	₹ 37,97,51,943.86	₹ 45,17,18,548.74
Operating Profit	₹ 1,45,37,500.00	₹ 14,36,62,530.00	₹ 17,08,22,283.50	₹ 22,03,48,445.64	₹ 28,28,04,328.01
Profit Margin	4.44%	35.87%	34.84%	36.72%	38.50%



- Application and Dashboard application development cost of INR 4Cr. In first year, INR 2Cr in second year
- Hardware Development cost is INR 2Cr in the first year
- Inflation of 5% per year
- Market share growth rate 20%
- Software expenses growth rate 15%
- Hardware expenses growth rate 10%
- Other expenses growth rate 15%
- Factory growth rate 2%
- Hardware cost per piece 20000

Exhibit-3: RICE Framework for Feature Prioritization

S.No	Features	Sub-Features	Reach	Impact	Confidence	Effort	Score
1	Customized digital checklist		4	4	4	1	64
2	Inspection dashboard	Availiability to track historic inspections, status update on scheduled inspections, trends	5	5	4	3	33
3	Incident management		5	4	4	3	27
4	Predict vulnerabe areas for inspection and recommend possible root cause and solutions		4	4	4	4	16
5	BLE for comunication and workforce tracking		4	4	3	4	12
6	AR based fault tagging		4	4	3	4	12
7	Central Repository to access historic data archives		4	2	3	2	12
8	Video based communication for remote assistance		4	3	4	4	12
9	Fault detection using image processing		4	3	4	5	9.6
10	Schedule an inspection event	Integrate with calender and notification sent to assigned personnel	2	2	2	1	8
11	Option to raise priority alerts	Call/one click high priority ticket creation for quick redressal	3	2	2	2	6
12	Speech-to-text to add details to an issue		3	2	3	3	6

Exhibit-4: Industry specific challenges

Oil & gas

- Very remote excavation centres so training on spot is costly and late in process
- Limited communication means so potential miscommunication, confusion with signs
- Costly to take an expert to remote site for small maintenance & repair tasks



Manufacturing

- Inventory management troublesome especially in post-pandemic world
- Machine diagnosis & repair tedious due to long waiting time for expert arrival
- Upskilling low-skilled workers needed

Mining

- Delay in alerting during natural calamities
- Poisonous gases detection
- Delay, inaccuracy in relaying SOS & location in emergency

Power

- Frequent technology upgrades necessitating continuous communication to avoid mishaps
- Only 23% of optimal training infrastructure present in India as per IEMR
- Competent management talent development needs adequate training due to highly technical staff

Chemicals

- Late fault detection due to irregular maintenance
- Manual check-list maintenance prone to human errors leading to fatal misses
- High risk due to improper labelling, faded markings

Exhibit–5 : SWOT Analysis

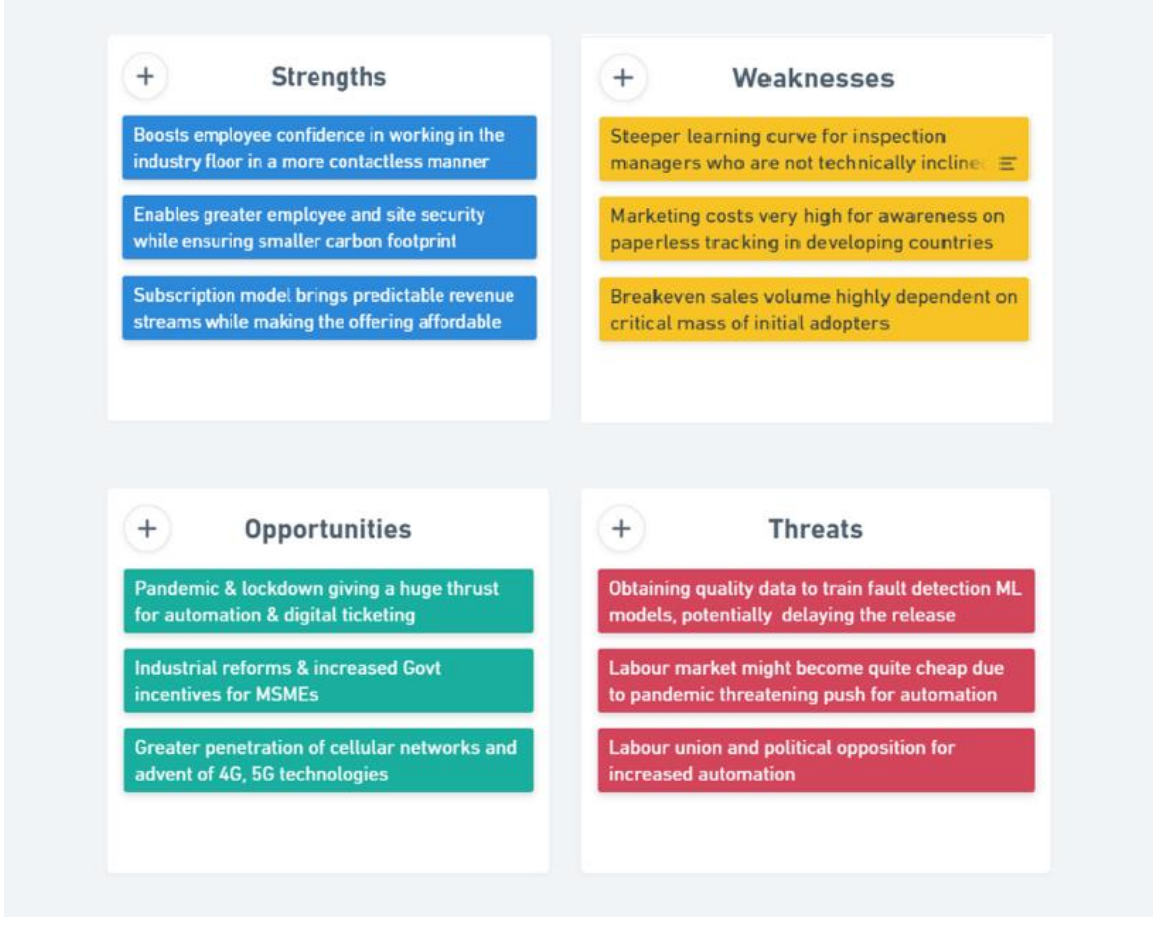
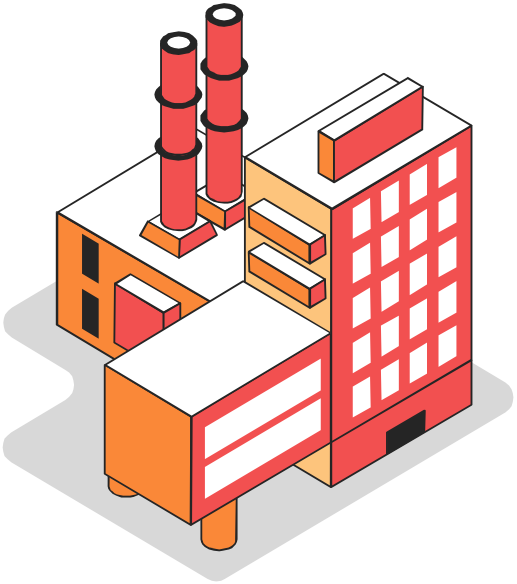


Exhibit-6 : References

HONEYWELL SWARM



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- https://www.honeywellprocess.com/en-US/explore/products/Pages/default.aspx
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- https://www.investindia.gov.in/sector/oil-gas
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